SCPC Homework - Exercise 2

Switched-Capacitor Power Converters

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Part (a)

Blocking Voltages of Power Switches

Part (a): Circuit A - Series-Parallel 4:1

Topology: 3 flying caps (C1, C2, C3), output cap C4, 10 switches

Conversion ratio: M = 1/4

Phase 1 (series): S1, S2, S3, S4 ON

Caps in series from Vin to Vout: Vin -> C1 -> C2 -> C3 -> Vout

Each cap voltage: Vc = Vin/4

Node voltages (phi1): [1, 3/4, 3/4, 1/2, 1/2, 1/4] x Vin

Phase 2 (parallel): S5-S10 ON

Each cap connected between Vout and GND

Node voltages (phi2): [1/4, 0, 1/4, 0, 1/4, 0] x Vin

Blocking voltage = |V_terminal1 - V_terminal2| when switch is OFF

Max blocking voltage: 3/4 x Vin (S1, S5, S8)

Part (a): Circuit A - Switch Blocking Voltages

Switch	Phase	Blocking Voltage	Switch	Phase	Blocking Voltage
S1	phi1	3/4 x Vin	S6	phi2	1/2 x Vin
S2	phi1	1/4 x Vin	S7	phi2	1/4 x Vin
S3	phi1	1/4 x Vin	S8	phi2	3/4 x Vin
S4	phi1	1/4 x Vin	S9	phi2	1/2 x Vin
S5	phi2	3/4 x Vin	S10	phi2	1/4 x Vin

Part (a): Circuit B - Ladder/Dickson 4:1

Topology: 3 flying caps (C1, C2, C3), output cap C4, 8 switches

Conversion ratio: M = 1/4

Cap voltages: V_C1 = V_C2 = Vin/2, V_C3 = Vin/4

Phase 1: S1, S2, S3 ON

Node voltages (phi1): [1, 1/2, 1/2, 1/4, 0, 1/4] x Vin

Phase 2: S4-S8 ON

Node voltages (phi2): [1, 0, 1, 0, 1/2, 1/4] x Vin

More uniform blocking voltage distribution than Circuit A

Most switches: 1/2 x Vin blocking voltage

Part (a): Circuit B - Switch Blocking Voltages

Switch	Phase	Blocking Voltage	Switch	Phase	Blocking Voltage
S1	phi1	1/2 x Vin	S5	phi2	1/4 x Vin
S2	phi1	1/2 x Vin	S6	phi2	3/4 x Vin
S3	phi1	1/2 x Vin	S7	phi2	1/2 x Vin
S4	phi2	1/4 x Vin	S8	phi2	1/4 x Vin

Part (b)

PMOS/NMOS Switch Implementation

Part (b): Switch Type Selection

Constraint: gate drive voltage in range [Vin, 0]

Selection rule:

Switches closer to Vin (high rail) -> PMOS

Gate driven to 0V to turn ON (Vsg = Vnode > |Vtp|)

Switches closer to GND (low rail) -> NMOS

Gate driven to Vin to turn ON (Vgs = Vin - Vnode > Vtn)

Circuit A (10 switches):

PMOS: S1, S5, S6, S7 (top rail switches, connected near Vin)

NMOS: S8, S9, S10 (bottom rail switches, connected to GND)

S2, S3, S4: intermediate nodes, either type possible

Circuit B (8 switches):

PMOS: S1, S2, S3, S5, S7, S8 (near high rail)

NMOS: S4, S6 (near low rail)

Part (c)

Ron, C1, C2, C3 Design for Rout = 20 Ohm

Part (c): Design Approach

Target: Rout = 20 Ohm at corner frequency fc = 10 MHz

Output impedance model:

$R_{out} = \sqrt{R_{SSL}^2 + R_{FSL}^2}$

At corner frequency: R_SSL = R_FSL

$R_{out} = \sqrt{2} \times R_{FSL} \Rightarrow R_{FSL} = 20/\sqrt{2} = 14.14$ Ohm

FSL (Fast Switching Limit):

$R_{FSL} = \sum(a_i^2) \times 2 \times R_{on}$ (switch charge multipliers)

SSL (Slow Switching Limit):

$R_{SSL} = m / (f_s \times C_T)$ (capacitor charge multipliers)

Capacitance density (for area):

$C_{1V8} = 6$ nF/mm^2 ($V_c < 1.8$V)

$C_{5V} = 2.5$ nF/mm^2 ($V_c < 5$V)

Part (c): Circuit A Sizing

Charge multiplier analysis:

FSL factor p = 5/4 => R_FSL = (5/4) x Ron

SSL factor m = 9/16

Ron calculation:

14.14 = (5/4) x Ron => Ron = 14.14 x 4/5 = 11.31 Ohm

Capacitor sizing:

R_SSL = m / (fs x CT) => CT = (9/16) / (10^7 x 14.14) = 3.977 nF

Equal distribution: C1 = C2 = C3 = CT/3 = 1.33 nF

Voltage across each cap: Vc = Vin/4 = 1V < 1.8V

Use C_1V8 density => Area = CT / 6 = 3.977/6 = 0.663 mm^2

Note: all caps are floating (C_INV type)

Part (c): Circuit B Sizing

Charge multiplier analysis:

FSL factor p = 17/8 => R_FSL = (17/8) x Ron

SSL factor m = 1

Ron calculation:

14.14 = (17/8) x Ron => Ron = 14.14 x 8/17 = 6.655 Ohm

Capacitor sizing:

R_SSL = m / (fs x CT) => CT = 1 / (10^7 x 14.14) = 7.07 nF

Optimal distribution: C3 = 2 x C1 = 2 x C2

C1 = C2 = 1.77 nF, C3 = 3.54 nF

All caps floating (C_INV), voltage < 1.8V for given Vin

Use C_1V8 density => Area = CT / 6 = 7.07/6 = 1.18 mm^2

Part (c): Circuit A vs Circuit B Comparison

Parameter	Circuit A	Circuit B
Topology	Series-Parallel	Ladder/Dickson
Ron	11.31 Ohm	6.655 Ohm
CT (total)	3.977 nF	7.07 nF
C1	1.33 nF	1.77 nF
C2	1.33 nF	1.77 nF
C3	1.33 nF	3.54 nF
Cap distribution	Equal	C3 = 2 x C1
Total area	0.663 mm^2	1.18 mm^2
Max blocking V	3/4 Vin	3/4 Vin

Part (d)

Maximum Theoretical Efficiency

Part (d): Efficiency Calculation

Given: Vin = 4V, Vout = 0.8V

Both circuits: M = 1/4 (ideal conversion ratio)

Maximum theoretical efficiency:

eta_max = Vout / (M x Vin)

eta_max = 0.8 / (1/4 x 4)

eta_max = 0.8 / 1.0

eta_max = 0.8 = 80%

Both Circuit A and Circuit B have the same ideal ratio M = 1/4

=> Same maximum theoretical efficiency: 80%

Note: actual efficiency will be lower due to:

- Switch conduction losses (R_on x I^2)
- Bottom-plate parasitic losses
- Gate drive losses
- Other parasitic effects

Part (e)

Circuit A Simulation: SSL vs FSL

Part (e): Simulation Setup

Circuit A only (series-parallel, M = 1/4)

Component values from part (c):

C1 = C2 = C3 = 1.33 nF, Cout = 80 nF, Ron = 11.31 Ohm

Two simulations at different switching frequencies:

Sim 1 - SSL regime (fs = fc/20 = 500 kHz):

Slow switching limit: fs << fc => Rout dominated by R_SSL

Iout reduced by 20x (0.5 mA) to keep Vout ~ M x Vin

Sim 2 - FSL regime (fs = 20fc = 200 MHz):

Fast switching limit: fs >> fc => Rout dominated by R_FSL

Iout = 10 mA (nominal load)

Part (e): SSL Results (fs = 500 kHz)

Rout ~ 282 Ohm (dominated by R_SSL ~ 1/fs)

Vout ~ 860 mV with large sawtooth ripple (~15 mV p-p)

Waveform characteristics:

Caps fully settle each phase (slow switching)

Charge transfers are impulsive: all charge dumps into Cout at once

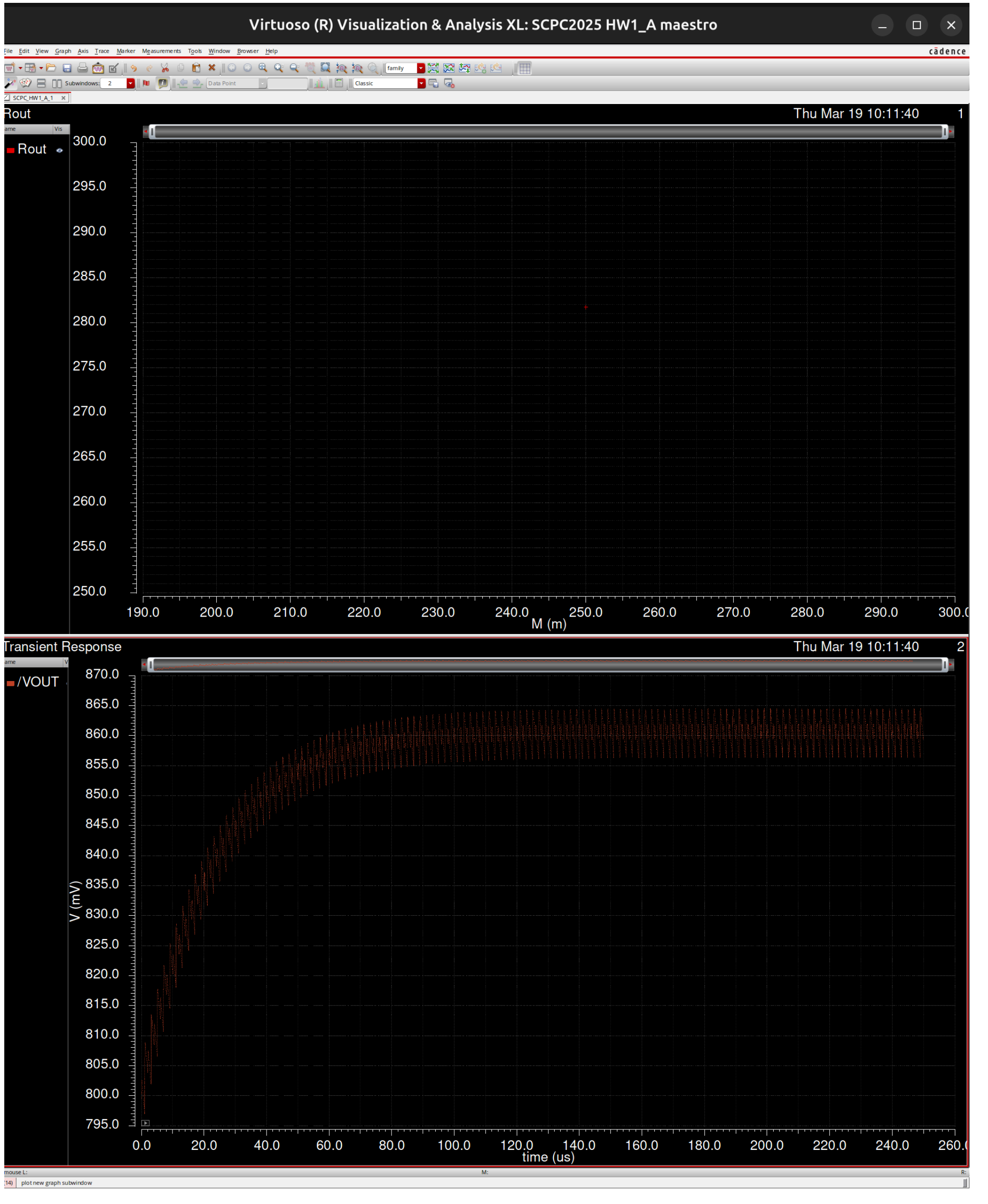
Causes visible voltage steps in output waveform

Iout reduced by 20x to keep Vout ~ M x Vin

because Vout = M x Vin - Rout x Iout

With 20x higher Rout, need 20x lower Iout to maintain Vout

Part (e): SSL Simulation (fs = 500 kHz)



SSL regime: Vout ~ 860 mV, large sawtooth ripple, Rout ~ 282 Ohm

Part (e): FSL Results (fs = 200 MHz)

Rout ~ 14.8 Ohm (dominated by R_FSL = p x Ron, independent of fs)

Vout ~ 852 mV with virtually no ripple (< 1 mV)

Waveform characteristics:

Caps barely change voltage each phase (fast switching)

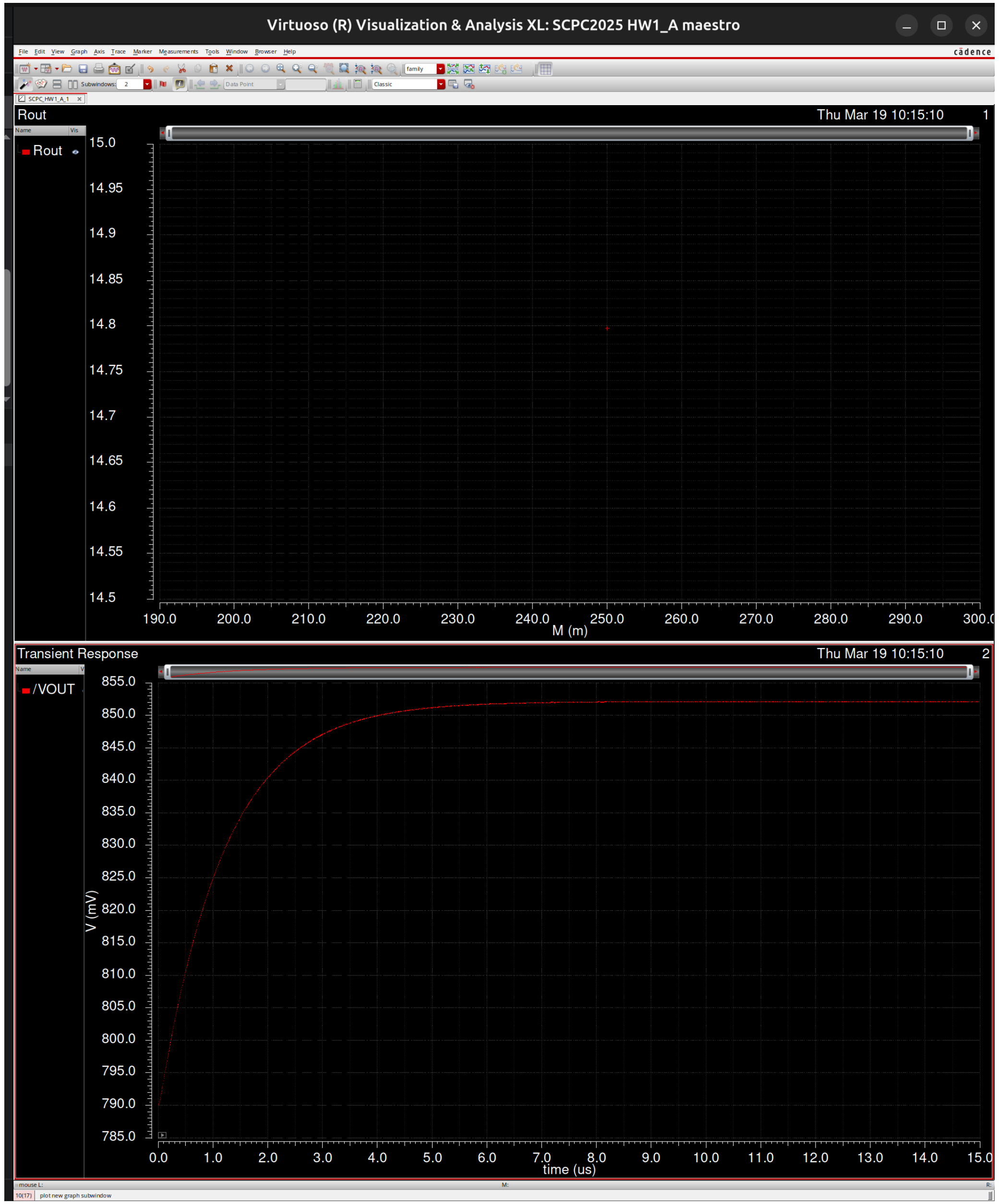
Current is quasi-constant => Cout sees smooth, continuous charging

Smooth exponential settling, no visible switching artifacts

At FSL, increasing fs further does NOT reduce Rout

Only reducing Ron can improve performance

Part (e): FSL Simulation (fs = 200 MHz)



FSL regime: Vout ~ 852 mV, virtually no ripple, Rout ~ 14.8 Ohm

Part (e): SSL vs FSL Key Takeaways

SSL (fs << fc): capacitance-limited losses

Rout ~ R_SSL = m / (fs x CT)

Increase C or fs to reduce Rout

Large sawtooth ripple from impulsive charge transfer

FSL (fs >> fc): resistance-limited losses

Rout ~ R_FSL = p x Ron (independent of fs)

Only reducing Ron helps

Negligible ripple from continuous charge transfer

Comparison:

SSL: Rout ~ 282 Ohm, ripple ~ 15 mV p-p

FSL: Rout ~ 14.8 Ohm, ripple < 1 mV

Fundamentally different charge transfer mechanisms:

SSL = impulsive (all charge at once) vs FSL = continuous (smooth current)

Part (f)

Bottom-Plate Parasitic Losses

Part (f): Bottom-Plate Loss Analysis

Given: Vin = 4V, caps sized as in part (c)

Parasitic capacitance:

Positive (top) plate: 0.1% of C

Negative (bottom) plate: 3% of C

Loss mechanism:

Parasitic caps charge/discharge as node voltages change between phases

P_bp = fs x sum(Cpar_i x delta_V_i^2)

Per-node parasitic (top + bottom plate at each node):

Cpar = C x (0.1% + 3%) = C x 3.1%

Key driver: node voltage swings between phi1 and phi2

Larger voltage swings at nodes => more bottom-plate loss

Circuit A has larger voltage swings at internal nodes

Part (f): Circuit A - Bottom-Plate Losses

Node voltage swings (normalized by Vin):

C1+: |1 - 1/4| = 3/4   C1-: |3/4 - 0| = 3/4

C2+: |3/4 - 1/4| = 1/2   C2-: |1/2 - 0| = 1/2

C3+: |1/2 - 1/4| = 1/4   C3-: |1/4 - 0| = 1/4

Sum of squared swings:

(3/4)^2 + (3/4)^2 + (1/2)^2 + (1/2)^2 + (1/4)^2 + (1/4)^2

= 9/16 + 9/16 + 4/16 + 4/16 + 1/16 + 1/16 = 28/16

Cpar per node = 1.33 nF x 3.1% = 41.23 pF

P_bp = 10 MHz x 6 nodes x Vin^2 x Cpar x 28/16

P_bp = 69.3 mW => more lossy

Part (f): Circuit B - Bottom-Plate Losses

Different parasitic caps (unequal C1,C2 vs C3):

Cpar_C1,C2 = 1.77 nF x 3.1% = 54.87 pF

Cpar_C3 = 3.54 nF x 3.1% = 109.7 pF

C1,C2 node voltage swing sum:

sum(delta_V^2) for C1,C2 plates = 7/16

C3 node voltage swing sum:

sum(delta_V^2) for C3 plates = 1

(C3+ swings full Vin, C3- stays at Vout)

P_bp = fs x Vin^2 x [7/16 x Cpar_C1C2 + 1 x Cpar_C3]

P_bp = 22.5 mW => less lossy

Circuit B has ~3x lower bottom-plate losses than Circuit A

Part (f): Bottom-Plate Loss Comparison

Parameter	Circuit A	Circuit B
C1 = C2	1.33 nF	1.77 nF
C3	1.33 nF	3.54 nF
Cpar (3.1%)	41.23 pF (all)	54.87 / 109.7 pF
Max node swing	3/4 x Vin	1 x Vin (C3+)
Sum(dV^2)	28/16	7/16 + 1
P_bp	69.3 mW	22.5 mW
	More lossy	Less lossy

Summary

Part (a): Blocking Voltages

Circuit A: max 3/4 x Vin (non-uniform distribution, 10 switches)

Circuit B: mostly 1/2 x Vin (more uniform, 8 switches)

Part (b): Switch Implementation

PMOS for switches near Vin, NMOS for switches near GND

Part (c): Component Sizing (Rout = 20 Ohm, fc = 10 MHz)

Circuit A: Ron = 11.31 Ohm, CT = 3.977 nF, Area = 0.663 mm^2

Circuit B: Ron = 6.655 Ohm, CT = 7.07 nF, Area = 1.18 mm^2

Part (d): Max Theoretical Efficiency

eta_max = 80% for both (M = 1/4, Vin = 4V, Vout = 0.8V)

Part (f): Bottom-Plate Losses

Circuit A: 69.3 mW (more lossy due to larger voltage swings)

Circuit B: 22.5 mW (less lossy, ~3x improvement)